How to make acrylic chess pieces with a laser cutter

by Steve Kranz
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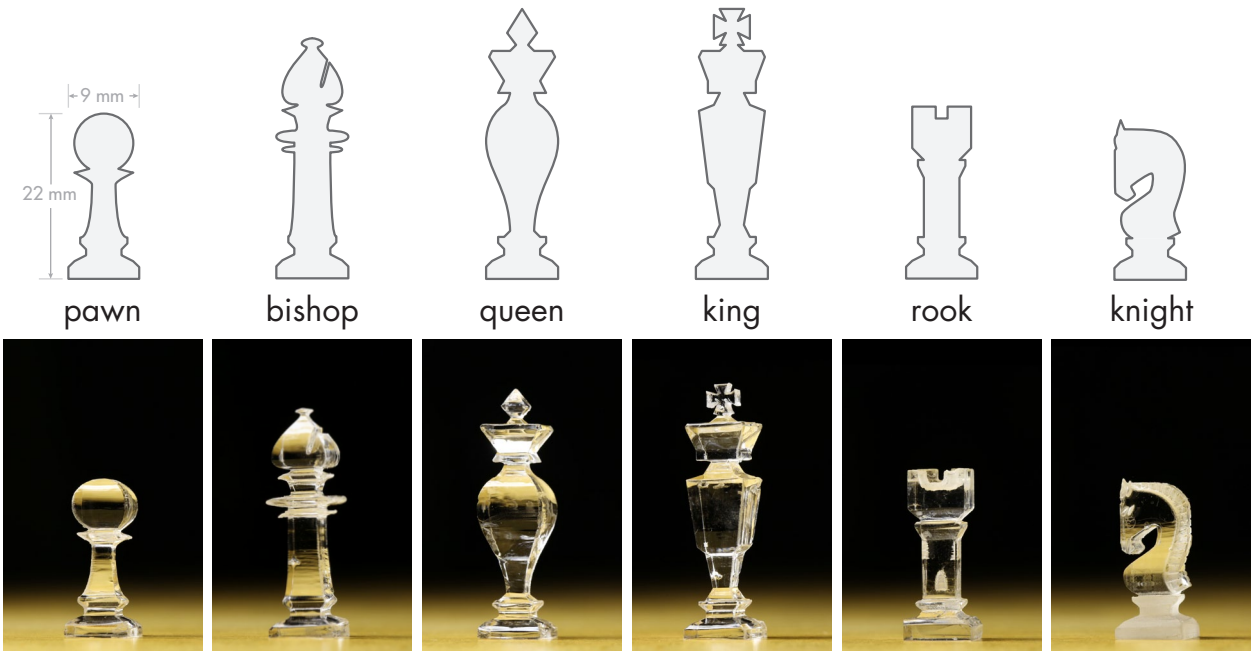


Required equipment	What I used
Laser cutter	Epilog Mini 18 (30 W) with 2" lens
Cast acrylic sheet/bar	3/8" x 2" x 8" *(9 mm x 50 mm x 200 mm)* **McMaster-Carr, #1227T339**
Vector drawing software	Adobe Illustrator

Time to make one piece

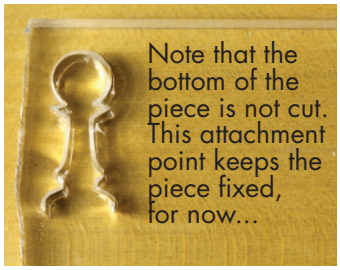
between 5 and 20 minutes
practice not included

Pawn

1. Cut the profile of the pawn piece.

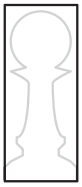
For a 30 W laser cutting through acrylic ~9 mm thick, I used settings of:
Power: 100%
Speed: 3%
Frequency: 5000 Hz

pawn Carve.pdf

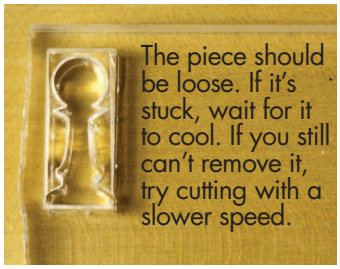


2. Cut a rectangle surrounding the piece.

All vector cuts will use the same power/speed/frequency settings as above.



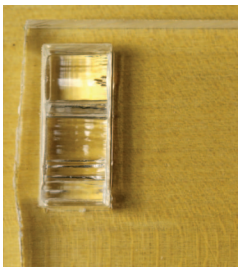
pawn Rect.pdf



3. Eject the piece along with the surrounding "padding."

4. Rotate the part 90° and place it back into the rectangular hole so that the top and bottom sides are now on the left and right.

At this point, we see that the piece should be as wide the material is thick.



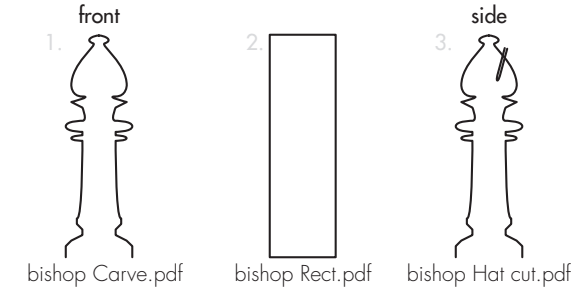
5. Cut the profile a second time.

pawn Carve.pdf

6. Remove the rest of the padding and enjoy your completed pawn!

Bishop

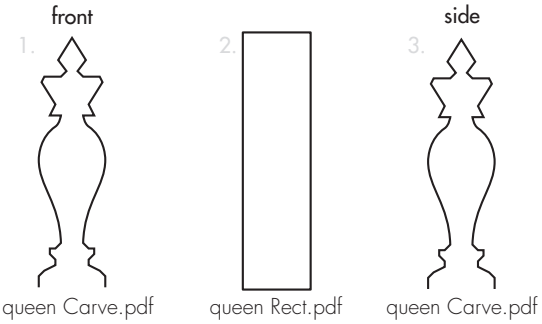
This bishop is made similar to the pawn. When cutting the side profile, cut a notch to help define the bishop's hat.



bishop Carve.pdf bishop Rect.pdf bishop Hat cut.pdf

Queen

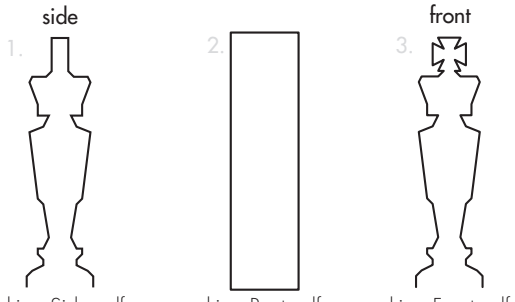
Like the pawn, the queen's front and side profiles are the same.



queen Carve.pdf queen Rect.pdf queen Carve.pdf

King

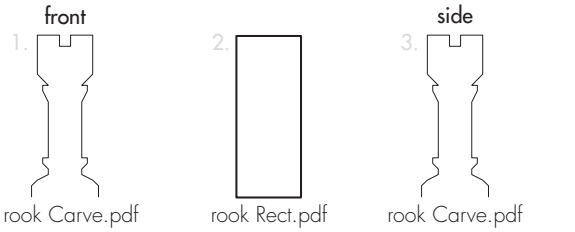
To form the cross on the king's crown, the side and front profiles are different. Theoretically, it doesn't matter which profile is cut first, but I found it easier to cut the side first.



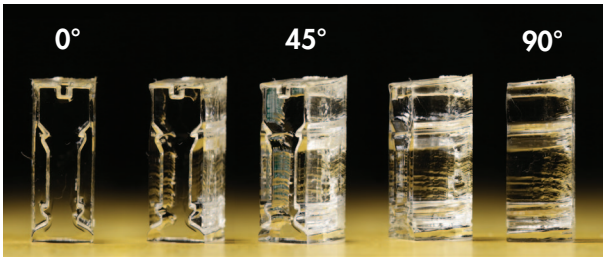
king Side.pdf king Rect.pdf king Front.pdf

Rook

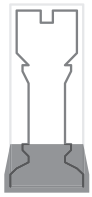
The rook is about as easy as the pawn to make. Both the front and side have the same profile.



rook Carve.pdf rook Rect.pdf rook Carve.pdf



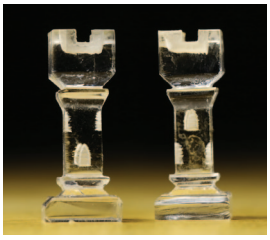
Instead of vector cutting, the piece can be etched to give the acrylic a frosted look.



rook Etch.pdf

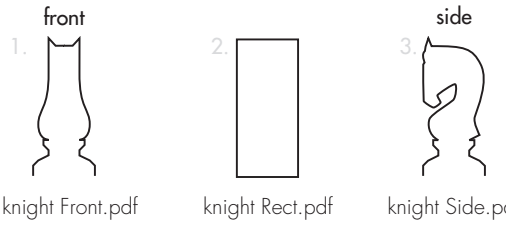
The same process can be used to make decorations.

rook Windows.pdf

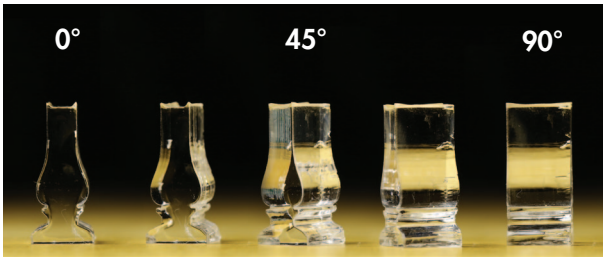


Knight

I think the knight is the most challenging, but most fun piece. The front and side profile are very different.

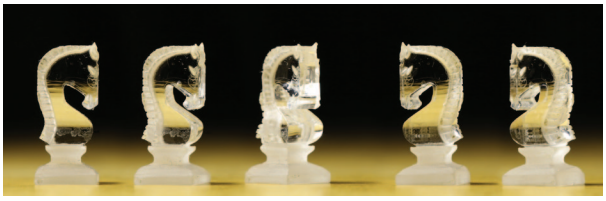


knight Front.pdf knight Rect.pdf knight Side.pdf



To help define the knight, etch eyes, nose, ears, mouth and a mane.

knight Face.pdf knight Mane1.pdf knight Mane2.pdf



Notes

Sometimes the "padding" sticks to the piece after a profile is cut. The vaporized acrylic can condense as a liquid and solidify between the piece and the padding. The edge of a razor blade or knife can pry the two apart.

Keep in mind that the laser beam has a finite width. In my case, around 0.2 mm. This means that if a laser cuts along a rectangle 9 mm wide and 30 mm tall, it yields a block that is 8.8 mm wide and 29.8 mm tall, and a hole that is 9.2 mm wide and 30.2 mm tall. Since the part is slightly smaller than the hole, it can wobble around and become misaligned when cutting the second profile *(pawn, step 5)*. To fix this, you could cut a seperate slightly smaller hole to fit the piece more snugly.

The laser beam cross-section is guassian—it is most intense in the center and drops off as it moves radially outward. As a result egdes cut by the beam have a slight angle. This can make the pieces stand not quite upright.

What if you made V-blocks that held the piece at 45° angles in the hole? Could you make an 8-sided chess piece?

Acknowledgements

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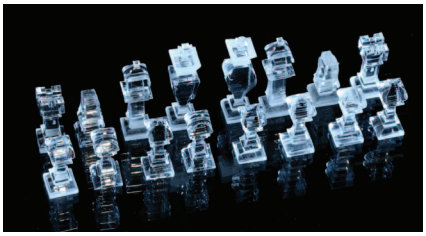

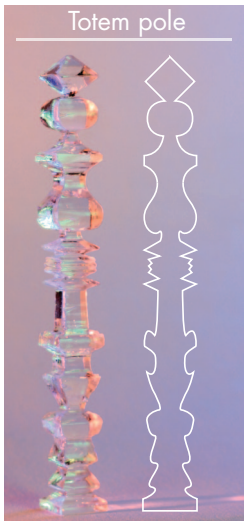

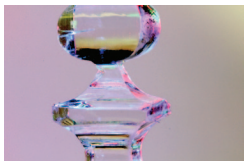

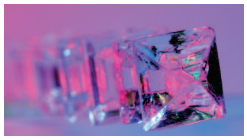

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